

Project Proposal (Proposed Solution) Report

Project Title: CaptionAI: AI-powered social media caption generator

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Maximum Marks: 5 Marks

The proposal report aims to transform social media content creation using advanced text processing models, boosting workflow efficiency and brand engagement. It tackles the severe creative block and operational inefficiencies faced by modern creators, promising streamlined marketing workflows, consistent posting schedules, and higher online visibility. Key features include an AI-driven prompt adaptation engine and real-time text layout formatting optimized for modern platform constraints.

Project Overview

- **Objective:** The primary objective is to revolutionize the copywriting workflow for solo business owners by implementing advanced language model techniques, ensuring faster, highly targeted, and expressive caption outputs.
- **Scope:** The project comprehensively assesses and enhances the digital content creation pipeline, incorporating generative AI capabilities for a more robust, agile, and cost-effective small business marketing strategy.

Problem Statement

- **Description:** Addressing the time-consuming inaccuracies, generic tones, and creative friction in the current manual caption drafting process that adversely affects operational efficiency and customer engagement frequency.
- **Impact:** Solving these issues will result in accelerated content delivery, reduced manual editing times, and an overall enhancement in organic reach, contributing to higher audience retention and brand scaling success.

Proposed Solution

- **Approach:** Employing prompt engineering and LLM integrations to analyze brief product notes and predict optimal marketing angles, creating a highly responsive and custom-tailored social media copy assistant.
- **Key Features:**
 - Implementation of a structured, context-aware prompt engine for tailored tone adjustment.
 - Real-time text generation and layout formatting matching specific character boundaries (e.g., Instagram/LinkedIn constraints).
 - Unified output blocks providing instant single-click caption, call-to-action (CTA), and hashtag packaging.

Resource Requirements

Resource Type PDF	Description PDF	Specification / Allocation PDF
Hardware		
Computing Resources	Core server specifications for deployment	AWS EC2 Virtual CPU Instances
Memory	RAM allocation for smooth server operations	4 GB
Storage	Storage allocation for user session logs and database data	20 GB SSD
Software		
Frameworks	Core application backend framework	Node.js (Express.js)
Libraries	UI formatting and model interaction dependencies	React.js, Tailwind CSS, Mongoose, Axios
Development Environment	Code editor and developer toolkit interface	VS Code (Visual Studio Code)
Data		
Data	AI training context parameters and layout properties	Platform constraint metadata, OpenAI Prompt payload context parameters